

Neural Networks

Seminar - Week 2 - Perceptron, MLP,
Activation functions and Forward pass

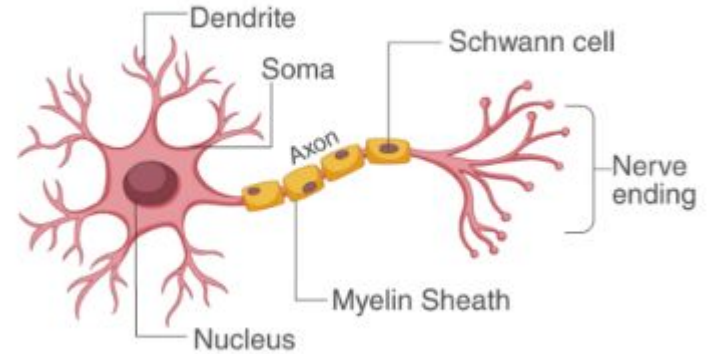
Assignment 1 Goals

- Implement vectorized linear layer as Perceptron
- Implement Multi-Layer Perceptron & forward pass
- Implement and explore activation functions
 - Sigmoid
 - TanH
 - ReLU
 - LeakyReLU

Artificial Neuron

Inspired by biological neuron

STRUCTURE OF NEURON



$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix}$$

Input feature vector

$$z = \sum_i^n w_i x_i + b$$

Linear
function

$$\hat{y} = g(z)$$

Activation
function

$$\hat{y}$$

Output
value

Perceptron

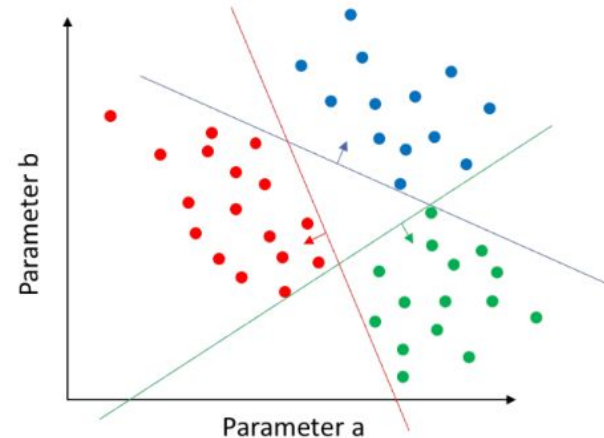
1958 - Rosenblatt - Discrete Perceptron

Step function - classification of linearly separable problems

A single artificial neuron with one output.

Parallel Combination of perceptrons -

Multiclass classification



Multi-Layer Perceptron

1985 - Rumelhart

Idea - Sequence of Single-Layer Perceptrons can learn complex patterns

Issue?

Multi-Layer Perceptron

1985 - Rumelhart

Idea - Sequence of Single-Layer Perceptrons can learn complex patterns

Issue - Sequence of linear functions is still a linear function

$$f(x) = ax + b$$

$$g(x) = cx + d$$

$$g(f(x)) = c(ax + b) + d$$

$$= acx + (cb + d)$$

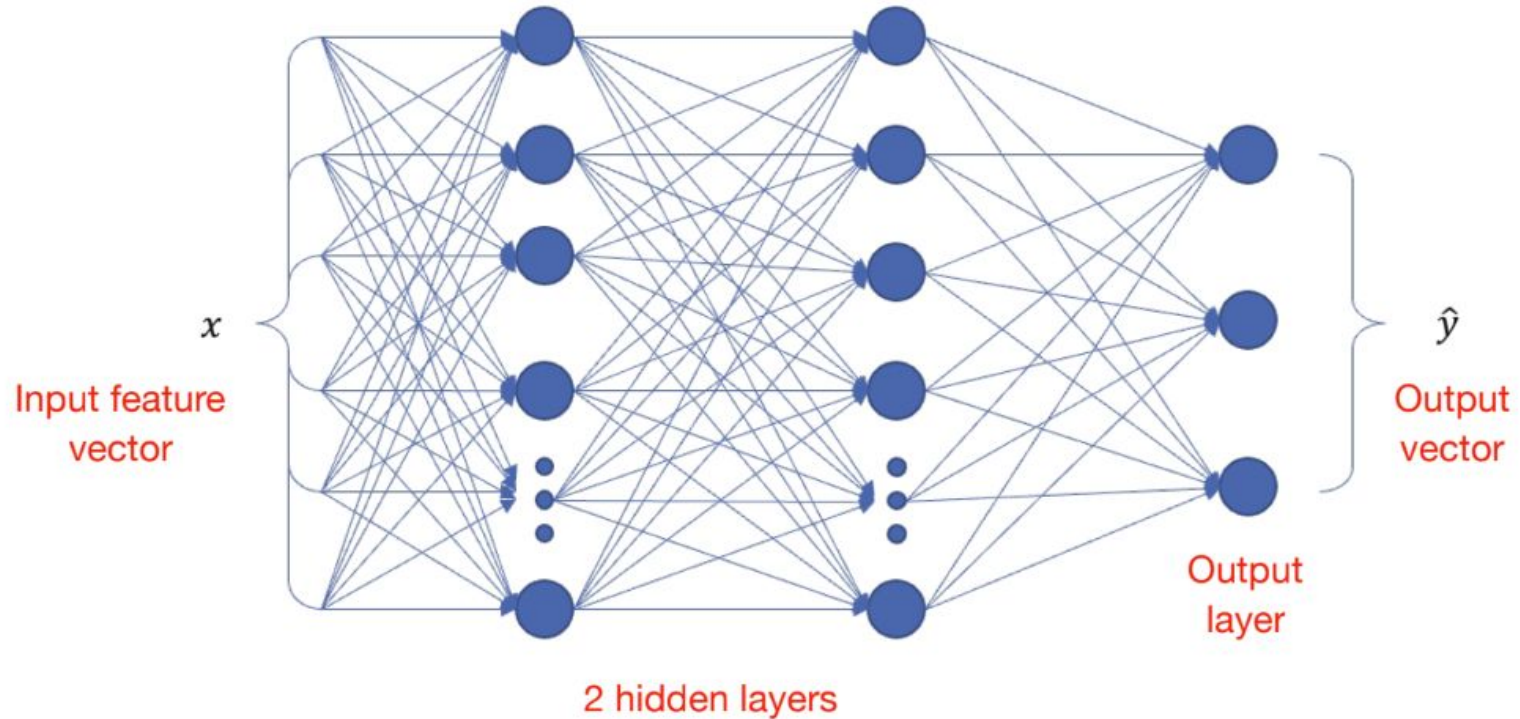
Linear function



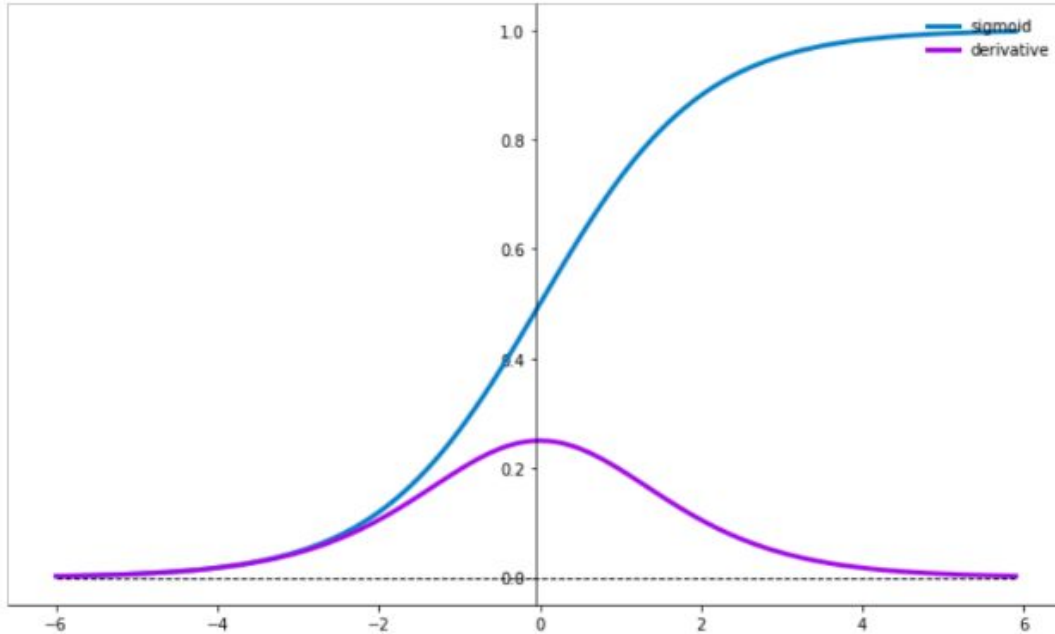
Multi-Layer Perceptron

A Neural Network Playground

Multi-Layer Perceptron



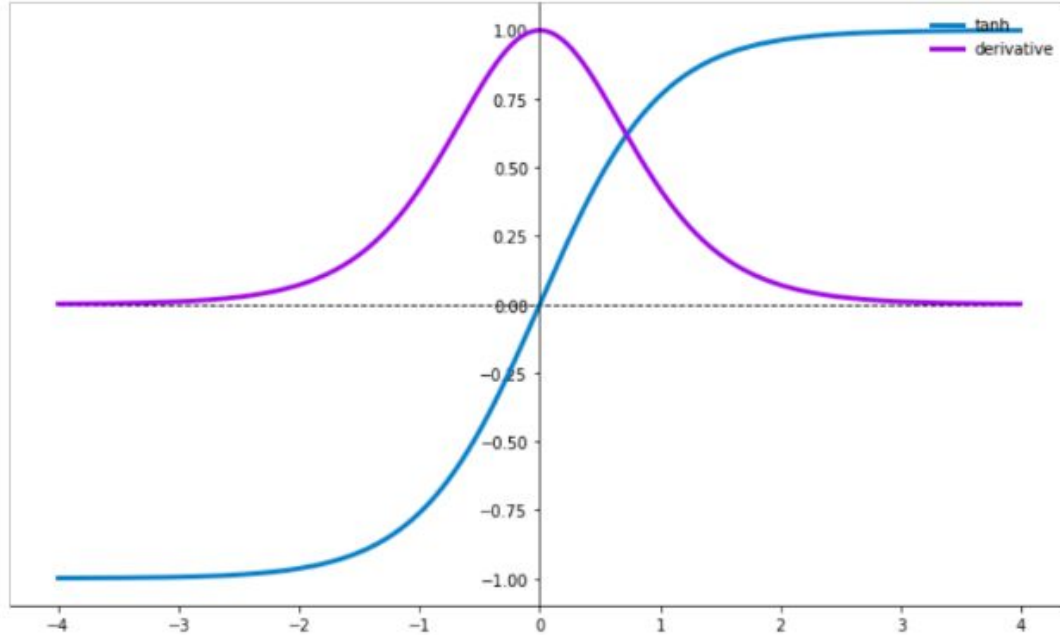
Activation Functions - Sigmoid



$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

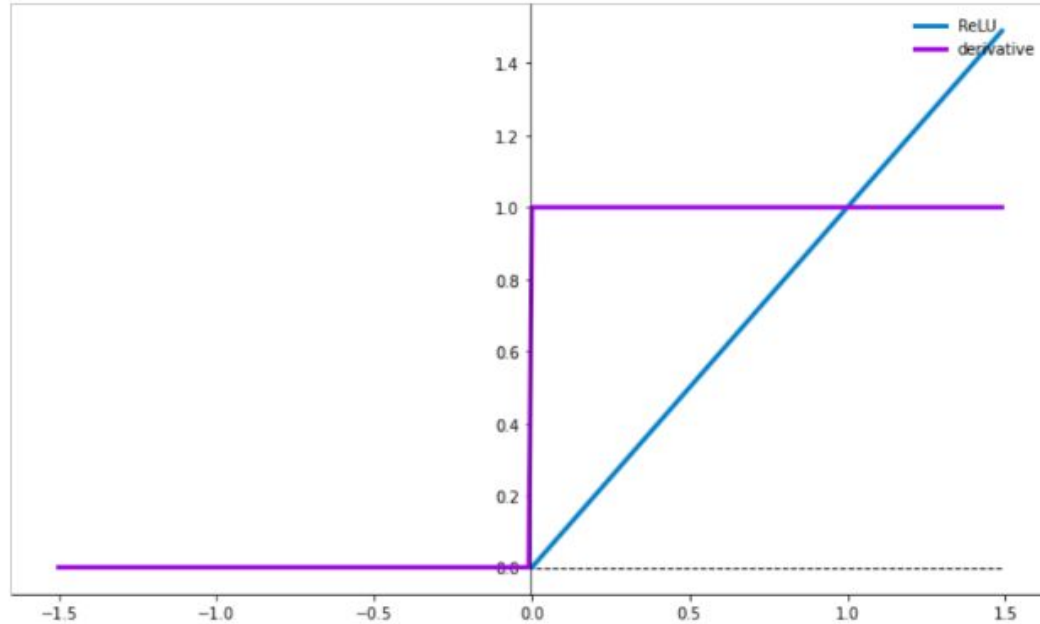
Activation Functions - TanH



$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$\tanh'(z) = 1 - \tanh^2(z)$$

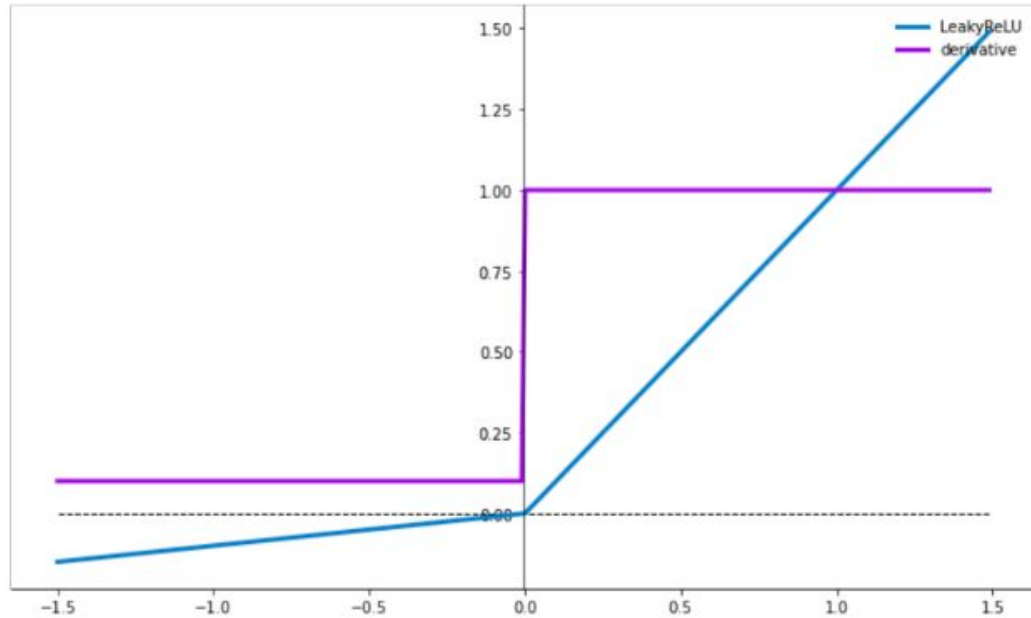
Activation Functions - ReLU



$$\text{ReLU}(z) = \max(0, z)$$

$$\text{ReLU}'(z) = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{else} \end{cases}$$

Activation Functions - LeakyReLU



$$\alpha = 0.1$$

$$lReLU(z) = \max(\alpha z, z)$$

$$lReLU'(z) = \begin{cases} 1 & \text{if } z > 0 \\ \alpha & \text{else} \end{cases}$$

Vectorized Form

Single unit - Single neuron

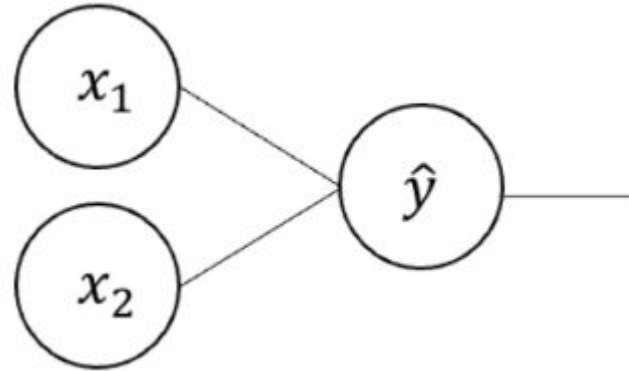
$$z = \sum_i^n w_i x_i + b$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$\hat{y} = a = g(z)$$

$$z = w^T x + b$$

$$z = \begin{pmatrix} - & w & - \end{pmatrix} \begin{pmatrix} | \\ x \\ | \end{pmatrix} + b$$



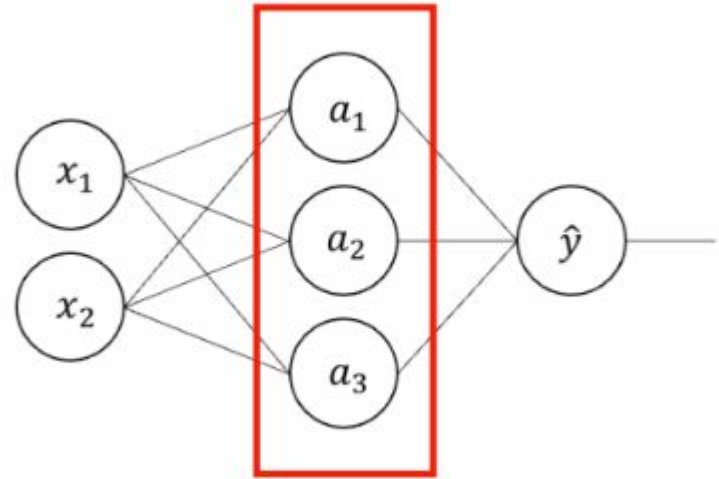
Vectorized Form

Hidden layer - Multiple neurons

$$\begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix} = \begin{pmatrix} - & w_1 & - \\ - & w_2 & - \\ - & w_3 & - \end{pmatrix} \begin{pmatrix} | \\ x \\ | \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$z = Wx + b$$

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = g\left(\begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}\right)$$



Vectorized Form

Multiple samples

$$\mathcal{X} = \{x^{(1)}, x^{(2)}, \dots, x^{(m)}\} \quad \mathcal{Y} = \{y^{(1)}, y^{(2)}, \dots, y^{(m)}\}$$

$$\begin{pmatrix} | & | & & | \\ z^{(1)} & z^{(2)} & \dots & z^{(m)} \\ | & | & & | \end{pmatrix} = \begin{pmatrix} - & w_1 & - \\ - & w_2 & - \\ - & w_3 & - \end{pmatrix} \begin{pmatrix} | & | & & | \\ x^{(1)} & x^{(2)} & \dots & x^{(m)} \\ | & | & & | \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$\mathbf{Z} = \mathbf{W}\mathbf{X} + \mathbf{b}$$

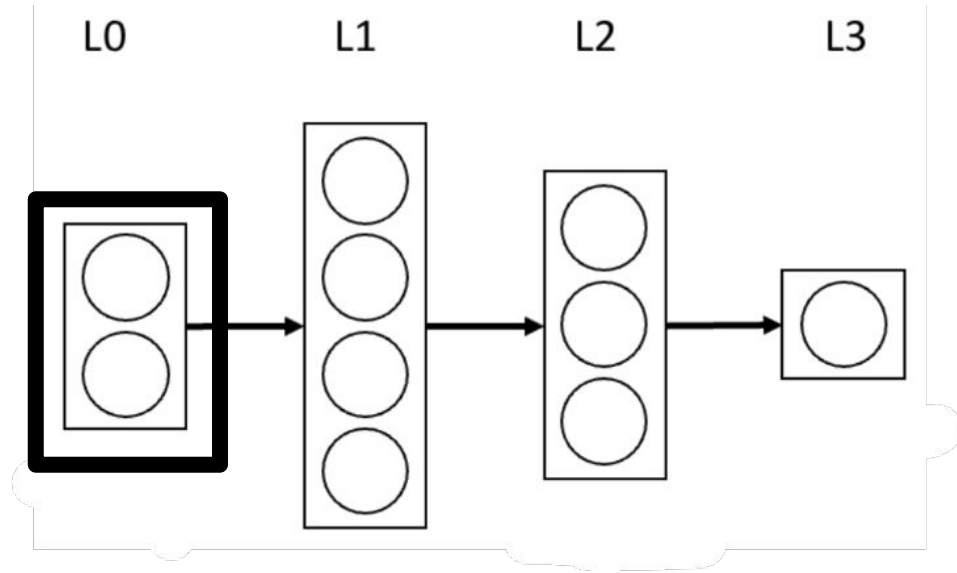
$$\mathbf{A} = g(\mathbf{Z})$$

Forward Pass

$$a^{[0]} = x$$

a - Activations

x - Input



Forward Pass

$$a^{[0]} = x$$

$$z^{[1]} = W^{[1]}a^{[0]} + b^{[1]}$$

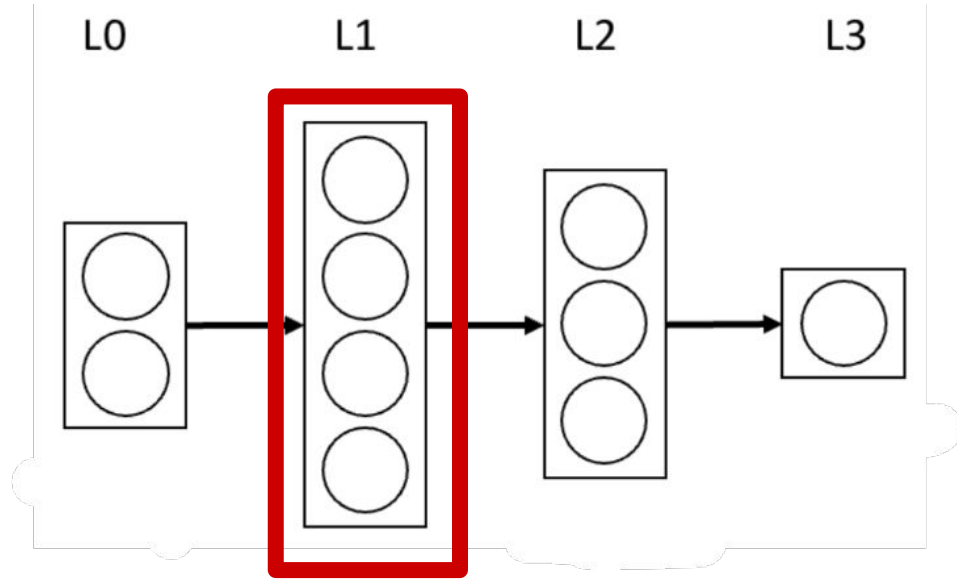
$$a^{[1]} = g^{[1]}(z^{[1]})$$

z - Pre-activation

W - Weights

b - Bias

g - Activation function



Forward Pass

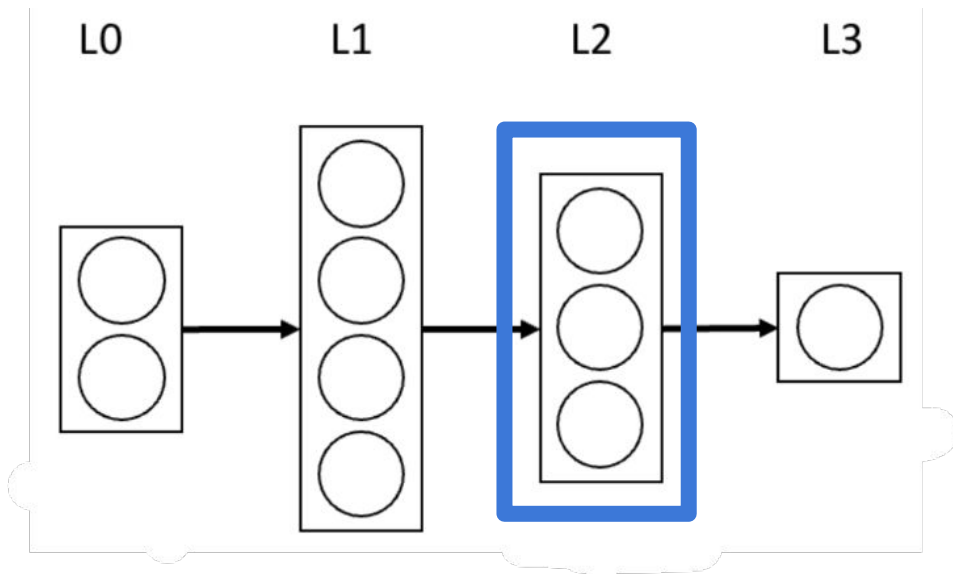
$$a^{[0]} = x$$

$$z^{[1]} = W^{[1]}a^{[0]} + b^{[1]}$$

$$a^{[1]} = g^{[1]}(z^{[1]})$$

$$z^{[2]} = W^{[2]}a^{[1]} + b^{[2]}$$

$$a^{[2]} = g^{[2]}(z^{[2]})$$



Forward Pass

$$a^{[0]} = x$$

$$z^{[1]} = W^{[1]}a^{[0]} + b^{[1]}$$

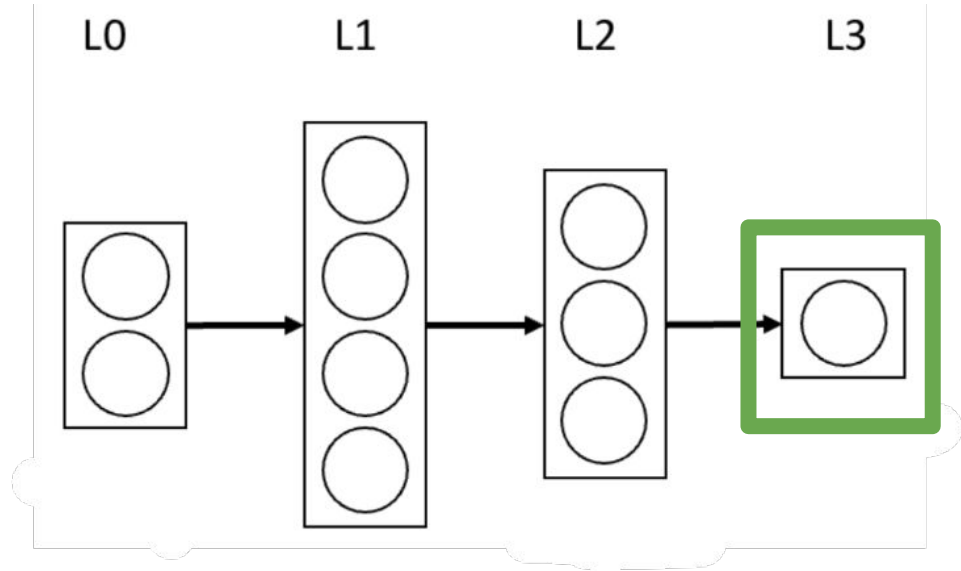
$$a^{[1]} = g^{[1]}(z^{[1]})$$

$$z^{[2]} = W^{[2]}a^{[1]} + b^{[2]}$$

$$a^{[2]} = g^{[2]}(z^{[2]})$$

$$z^{[3]} = W^{[3]}a^{[2]} + b^{[3]}$$

$$a^{[3]} = g^{[3]}(z^{[3]})$$



Week 3 - TASK

- Implement vectorized linear layer as Perceptron
- Implement Multi-Layer Perceptron & forward pass
- Implement and explore activation functions

https://github.com/vgg-fiit/NSIETE_2026